Track 5: Robotics and Automation

This track encourages the design of machines that reduce repetitive, physically demanding, or hazardous agricultural work. Solutions must cope with uneven terrain, unstructured environments, and limited budgets, and should aim for reliability and safety around people.

Problem Title 1: Autonomous Agricultural Weeding Robot

A. Problem Description

Weeds compete with crops for water, nutrients, and light, and weeding is among the most labour-intensive tasks on many farms. Manual weeding is slow and costly, and indiscriminate herbicide use raises cost and environmental concerns.

A robot that moves along crop rows, recognises weeds, and removes or treats them selectively could lower labour needs and chemical use.

B. Challenges

1. Distinguishing weeds from crop plants under varying light, growth stage, and soil conditions.
2. Navigating uneven, muddy, or crowded rows reliably.
3. Designing a weeding mechanism that is effective yet gentle on crops.
4. Battery life, weight, and soil compaction.
5. Keeping cost low and the machine easy to maintain.

Problem Title 2: Automated Fruit and Vegetable Harvesting Robot

A. Problem Description

Harvesting requires labour at specific times, and shortages of workers can lead to delayed harvest and losses. Produce must be picked at the right maturity and handled gently to avoid bruising.

A robotic picker that identifies ripe produce and picks it carefully could reduce dependence on scarce labour.

B. Challenges

1. Detecting fruit that is hidden by leaves, overlapping, or in variable light.
2. Assessing maturity reliably from colour, size, or other features.
3. Positioning the arm accurately and planning motion around plants.
4. Designing grippers that hold produce securely without damage.
5. Achieving acceptable speed and cost compared with manual picking.

Problem Title 3: Smart Seed Sowing and Precision Planting Robot

A. Problem Description

Uneven seed spacing and depth lead to patchy germination, competition between plants, and wasted seed. Manual sowing is slow, and simple mechanical seeders may not adjust to different seeds or soil conditions.

A precise planting robot can place seeds consistently and record where they were sown.

B. Challenges

1. Metering different seed sizes and shapes accurately without clogging or damage.
2. Controlling depth on uneven or hard soil.
3. Maintaining straight, accurate travel and spacing.
4. Detecting missed or doubled seeds.
5. Adapting to different crops and keeping the cost low.

Problem Title 4: Autonomous Agricultural Material Transport Robot

A. Problem Description

Carrying harvested produce, inputs, and tools across fields and plantations is strenuous and repetitive, particularly on slopes and in wet conditions. This work takes labour from tasks needing greater skill.

A load-carrying robot that follows a worker or a defined route can reduce physical strain and save time.

B. Challenges

1. Traction and stability on slopes, mud, and uneven ground.
2. Safe navigation among people, animals, and obstacles.
3. Load capacity relative to battery life and cost.
4. Localisation without reliable GPS under tree cover.
5. Rugged, weather-resistant construction.

Problem Title 5: Automated Fertilizer and Manure Application Robot

A. Problem Description

Manual spreading of fertilizer and manure is tiring, and application is often uneven, with some plants receiving too much and others too little. Over-application wastes money and can harm soil and water, while handling manure is unpleasant and unhealthy.

A robot that delivers measured quantities close to plants can improve consistency and reduce manual handling.

B. Challenges

1. Accurate metering of materials with different forms, such as granules, powders, and moist manure.
2. Preventing clogging and corrosion in dispensing mechanisms.
3. Determining application quantities from crop and soil information.

4. Navigating rows accurately and positioning the dispenser near plants.
5. Handling payload weight, spills, and operator safety.

Problem Title 6: Automated Agricultural Sorting and Grading Robot

A. Problem Description

Manual sorting is slow and subjective, and grading differs between workers and over a working day. Inconsistent grading affects the price farmers receive and the quality that buyers expect.

An automated system that measures size, colour, shape, or visible defects can grade produce more consistently and speed up processing.

B. Challenges

1. Defining clear, measurable grading criteria that match market standards.
2. Obtaining images of defects and grades for training and testing.
3. Maintaining consistent lighting and handling items individually at useful speed.
4. Designing reliable mechanisms to divert items into grades without damage.
5. Calibrating and maintaining the system at low cost.